Evaluating AI Model Performance Using a Simulated Human-in-the-Loop System

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1. Project Overview

This project simulates an AI model evaluation workflow using a structured dataset created in Google Sheets. The analysis focuses on how different model versions perform across prompt types, as well as how human annotators influence evaluation quality and efficiency. The goal was to identify performance gaps, error patterns, and operational inefficiencies in a human-in-the-loop system.

The dataset was intentionally designed to reflect real-world AI evaluation patterns, including model predictions, ground truth labels, confidence scores, and rework requirements. In addition to model performance, this project explores the role of human-in-the-loop systems by analyzing annotator efficiency, evaluation time, and the impact of rework on overall workflow performance.

Through this analysis, the project highlights key challenges in AI evaluation, including inconsistent model accuracy, error patterns such as bias and hallucination, and the operational cost of maintaining quality through human review.

Executive Snapshot

- **Overall model accuracy:** ~82% across 300 evaluations
- **Weakest performance area:** Safety prompts (~77% accuracy), indicating higher-risk scenarios
- **Most common errors:** Bias and hallucination, suggesting issues with content reliability
- **Annotator insight:** Junior annotators achieved the highest accuracy, while senior annotators were slightly faster

2. Business Problem

AI systems are only as reliable as their outputs—and their evaluation processes.

Organizations need to answer:

- Can we trust this model's predictions?
- Where does the model fail most often?
- How much human effort is required to maintain quality?

This project simulates these challenges and analyzes both **model performance** and **human-in-the-loop efficiency**.

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3. Dataset Description

The dataset contains 300 simulated AI evaluation records generated using Google Sheets formulas.

Key Features:

- Model versions: v1.0, v1.1, v2.0
- Prompt types: Informational, Creative, Safety, Technical
- Ground truth vs model predictions
- Annotator levels: Junior, Mid, Senior
- Evaluation time, confidence score, and rework indicators

Data Design Notes:

- Approx. 80% model accuracy
- Errors include hallucination, bias, false positives, false negatives
- Rework is correlated with incorrect predictions

- Designed to mimic real-world AI evaluation workflows

Metric	Value	Why It Matters
Overall Accuracy	82.0%	Baseline model performance across all evaluations
Best Prompt Type	Informational – (89%)	Strong performance on factual queries
Worst Prompt Type	Safety – (77%)	Highest-risk area requiring improvement
Average Confidence	49%	Indicates misalignment between confidence and accuracy

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4. Methodology

The analysis was conducted using Google Sheets.

Techniques Used:

- Pivot tables for aggregation
- Calculated fields (accuracy, averages, percentages)
- Data grouping (confidence buckets)
- Charts for visualization

Key Metrics:

- Accuracy rate

- Rework rate
- Average evaluation time
- Confidence vs accuracy relationship

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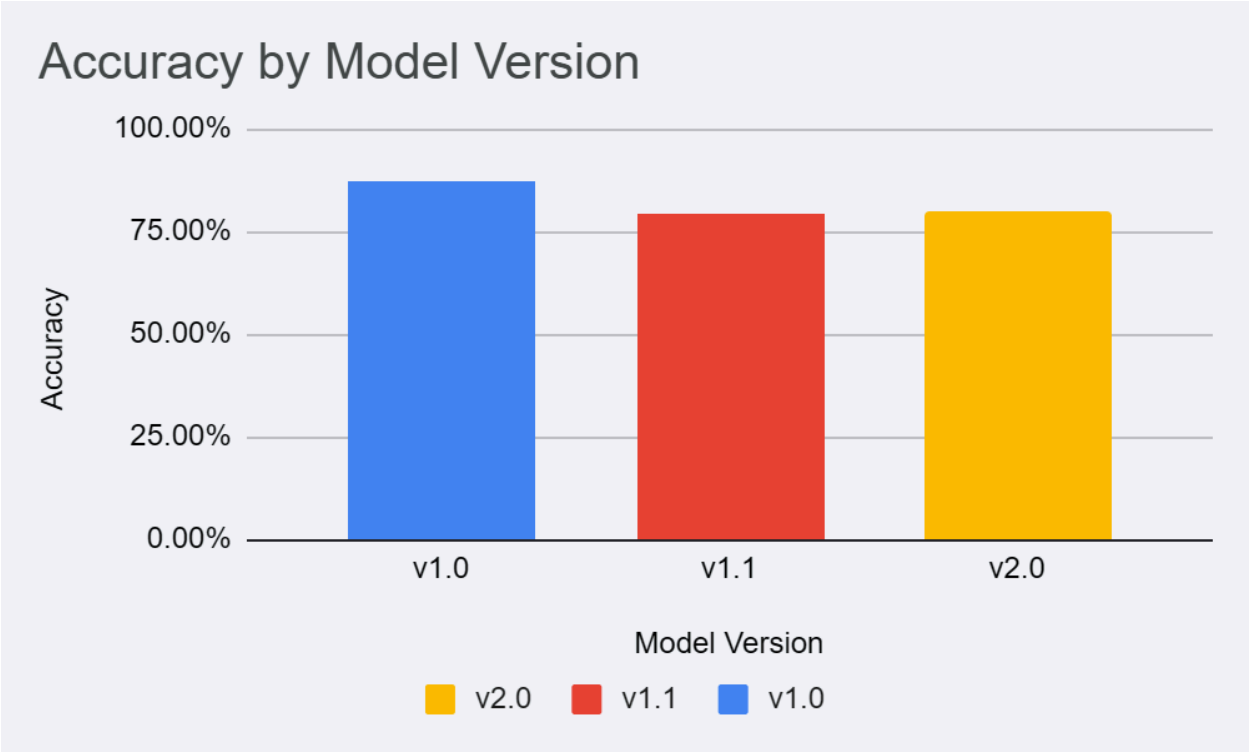
5. Key Findings

5.1 Model Accuracy by Version

Insight:

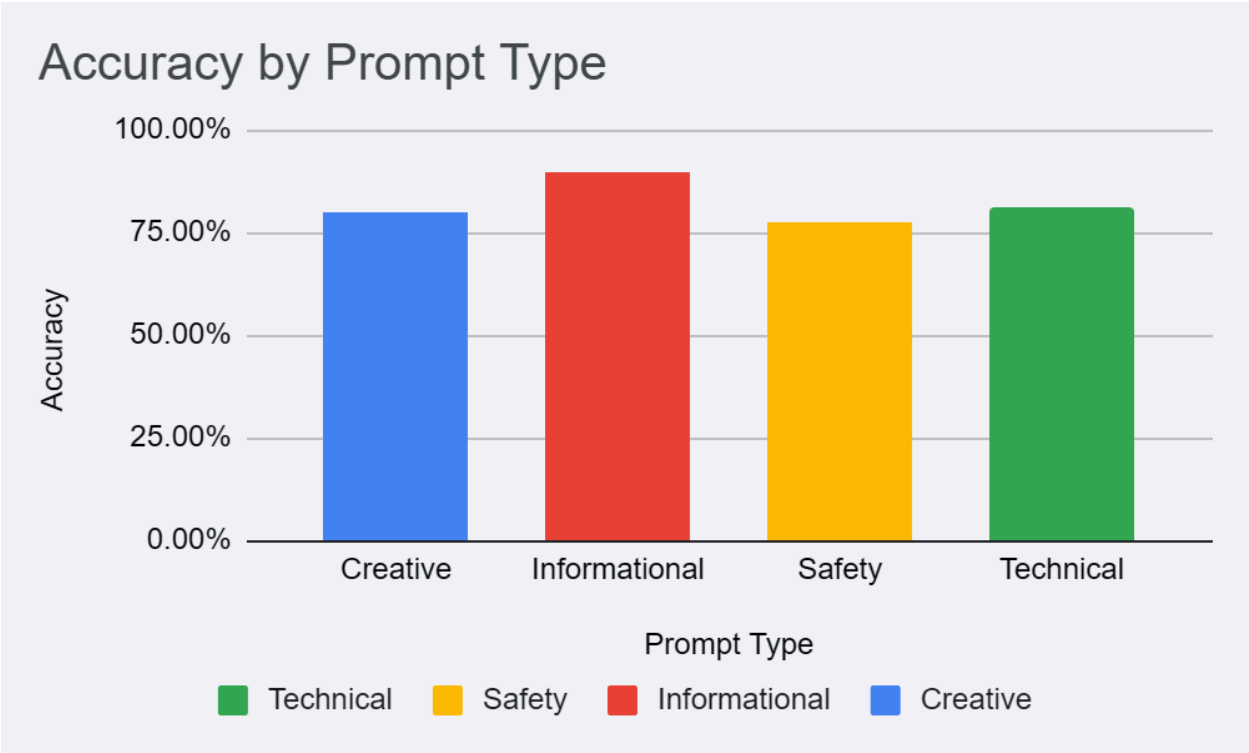
Model performance varied across versions, with v1.0 achieving the highest accuracy (~87%), while v1.1 and v2.0 showed lower and more consistent performance (~79–80%). This suggests that newer versions did not significantly improve overall accuracy and may require further optimization.



5.2 Accuracy by Prompt Type

Insight:

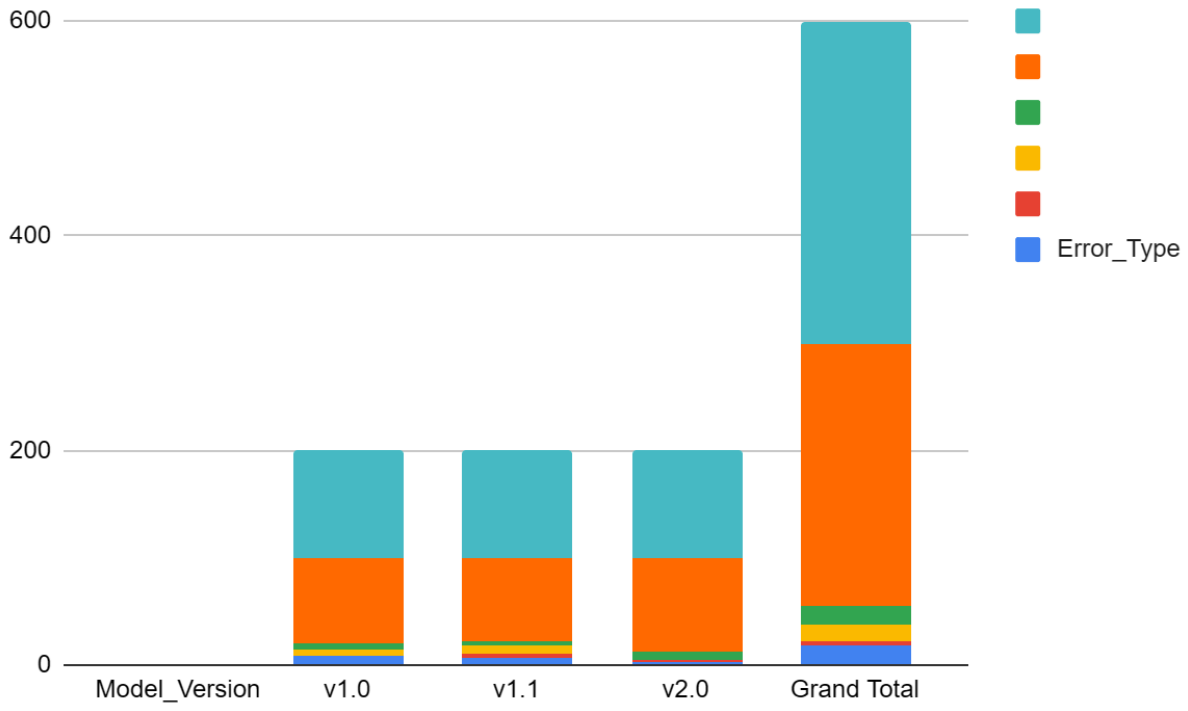
Model performance varied significantly by prompt type. Informational prompts achieved the highest accuracy (~89%), while safety-related prompts had the lowest accuracy (~77%). This indicates that the model performs well on straightforward, factual queries but struggles with more complex or sensitive content, highlighting a key area for improvement.



5.3 Error Type Distribution

Insight:

Bias and hallucination were the most frequently occurring error types, while false positives and false negatives were less common. This suggests that the model's primary weaknesses are related to generating inaccurate or misleading content rather than simple classification mistakes, indicating a need for improved training data and evaluation in these areas.



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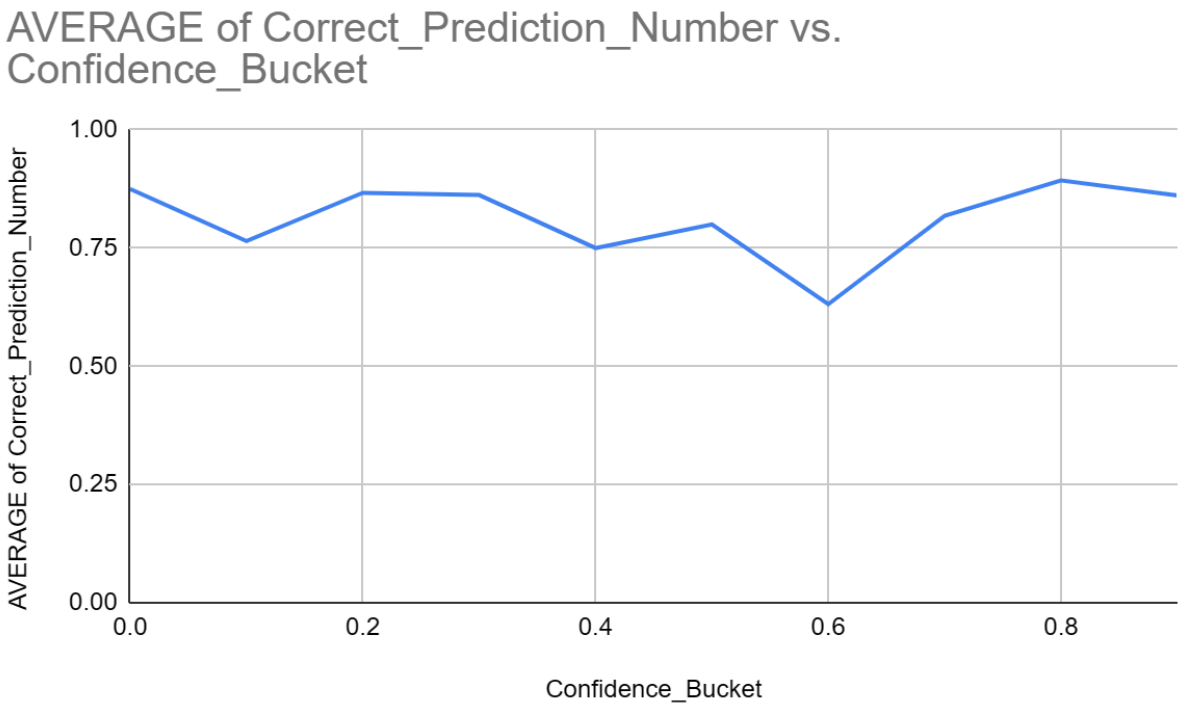
6. Advanced Analysis

6.1 Confidence vs Accuracy

Insight:

Model confidence does not show a consistent relationship with prediction accuracy. While higher confidence scores occasionally align with better performance, the overall trend is inconsistent, indicating that confidence is not a reliable indicator of correctness.

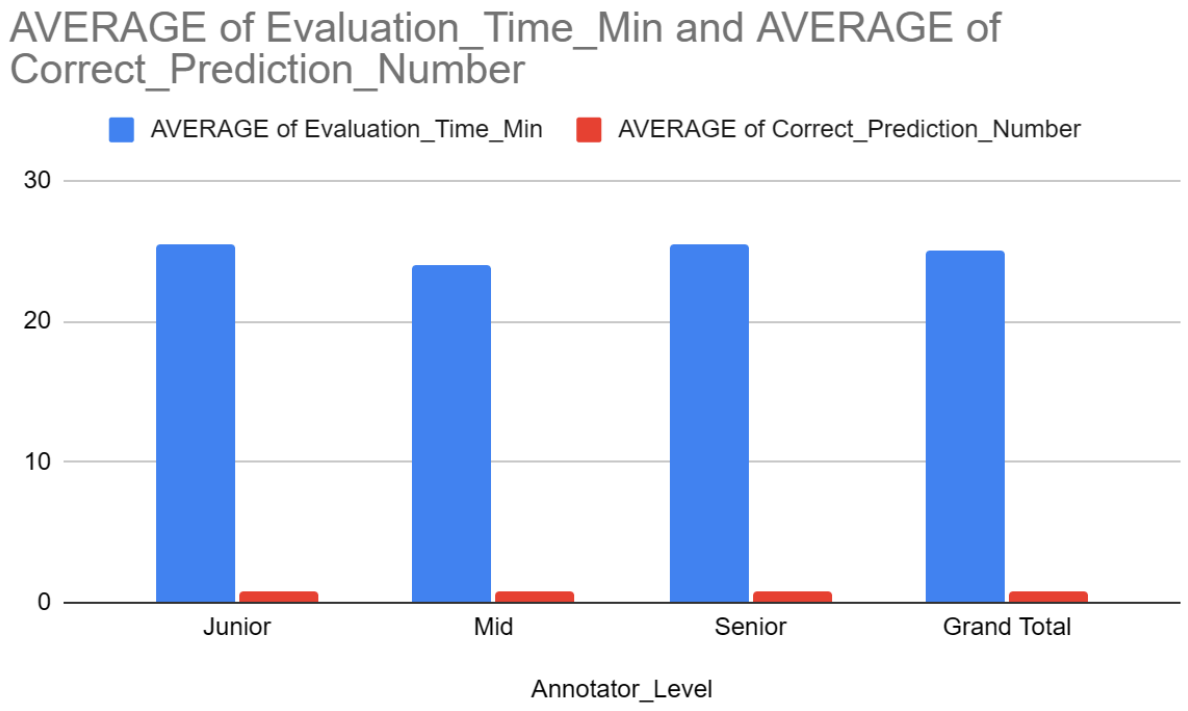
Critical Finding:
At a confidence level of 1.0, model accuracy dropped to approximately 50%, indicating cases where the model was **"hallucinating with certainty."** This represents a significant reliability risk, as high confidence is often interpreted as trustworthiness. This finding highlights the need for improved confidence calibration to better align model certainty with actual performance.



6.2 Annotator Efficiency

Insight:
Annotator performance showed minimal differences in evaluation time across experience levels, with senior annotators being slightly faster on average. However, junior annotators achieved the highest accuracy (~87%), outperforming both mid-level and senior annotators (~79–80%).

This suggests that increased experience does not necessarily lead to higher accuracy in this workflow. While senior annotators may complete evaluations more quickly, junior annotators may be more cautious, resulting in more accurate outcomes. This highlights a potential tradeoff between speed and accuracy in human-in-the-loop systems.



6.3 Rework Impact

Insight:
Rework cases accounted for a smaller portion of total evaluations but still contributed additional operational effort. While individual rework evaluations showed a slightly lower average evaluation time, this is likely due to their lower frequency rather than increased efficiency.

The presence of rework indicates that initial model predictions or evaluations were insufficient, requiring additional human intervention. Even at lower volumes, rework introduces inefficiencies and increases the total time required to maintain output quality. Reducing rework would improve overall system efficiency and reduce unnecessary human effort.

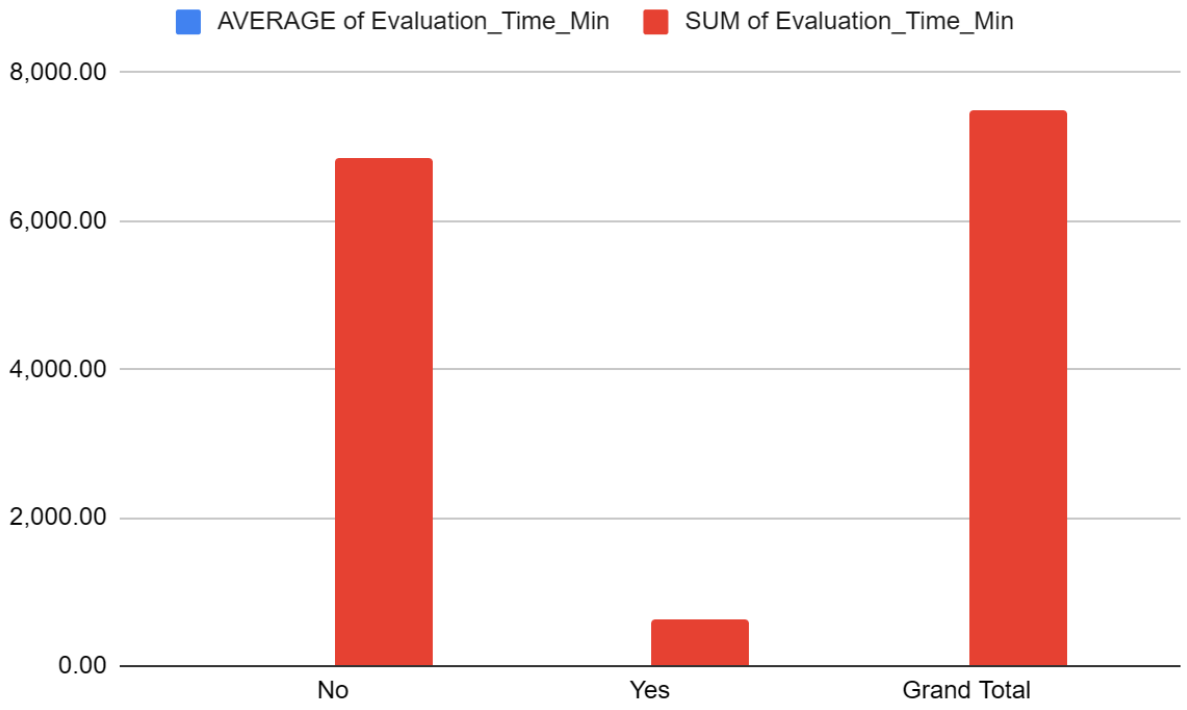
6.4 Error Behavior by Model Version

Insight:
Error patterns varied across model versions, indicating differences in how each model fails.

Model v2.0 demonstrated the strongest overall performance, with the highest number of correct predictions (“None” errors) and fewer total errors compared to earlier versions.

Model v1.1 showed increased rates of false positives and false negatives, suggesting instability in classification tasks. In contrast, model v1.0 exhibited higher instances of bias and hallucination, indicating a tendency toward generating inaccurate or misleading content.

These differences highlight that model improvements are not only reflected in overall accuracy but also in the types of errors produced. Understanding these patterns is critical for targeted model optimization and risk mitigation.



6.5 Resource Impact

Insight:
Annotator performance suggests a tradeoff between speed and accuracy, with senior annotators completing tasks slightly faster and junior annotators achieving higher accuracy. This indicates that an optimized staffing model could improve overall efficiency by assigning complex

or high-risk evaluations to more accurate annotators while leveraging faster annotators for simpler tasks.

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7. Recommendations:

- The model should improve performance on safety prompts by implementing stricter bias detection and enhanced error-handling procedures. This is necessary because safety-related prompts showed the lowest accuracy and were more frequently associated with bias-related errors and misclassifications. Strengthening these areas would improve reliability and reduce risk in sensitive use cases.

- The model's confidence scoring system should be improved by implementing better calibration techniques to ensure that confidence levels more accurately reflect prediction accuracy. The analysis showed that confidence scores were inconsistent and did not reliably align with correct outcomes, indicating that the model may be overconfident in certain predictions. Improving calibration would increase trust in the model's outputs and reduce the risk of relying on misleading confidence signals.

- The evaluation workflow should be optimized by aligning task assignments with annotator strengths and balancing speed with accuracy. The analysis showed that junior annotators achieved higher accuracy, while senior annotators completed evaluations slightly faster. This suggests a tradeoff between speed and quality. Assigning more complex or high-risk evaluations to more accurate annotators, while leveraging faster annotators for simpler tasks, could improve overall efficiency and output quality
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- Rework should be minimized by improving initial prediction accuracy and strengthening early-stage evaluation processes. The analysis showed that although rework cases were less frequent, they still contributed to additional operational effort and inefficiency. Reducing the need for rework would decrease total evaluation time, lower human workload, and improve overall system efficiency. This could be achieved through better model training, clearer evaluation guidelines, and earlier detection of incorrect predictions.

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8. Limitations

This project is based on a simulated dataset designed to reflect real-world AI evaluation patterns. While the structure and logic were intentionally created to mimic realistic scenarios, the results may not fully capture the complexity and variability of real production environments.

Additionally, assumptions were made when estimating factors such as rework impact and evaluation efficiency. The dataset also does not include real user interaction data, which may influence model behavior and error patterns in practice.

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9. Next Steps

Future work could expand this analysis by incorporating more diverse and complex datasets to better reflect real-world AI evaluation scenarios. Integrating real user interaction data would provide deeper insight into model performance and behavior under practical conditions.

Additionally, building automated dashboards could improve monitoring and reporting efficiency, while further exploration of model retraining strategies could help address identified weaknesses such as bias, hallucination, and inconsistent confidence scoring.

Final Insight

This analysis demonstrates that improving AI systems is not only about increasing accuracy, but about understanding how models fail, how humans interact with them, and how those factors impact real-world reliability and risk